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The application of biosecurity practices for preventing avian influenza in North-Eastern Italy turkey farms: An analysis of the point of view and perception of farmers

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ABSTRACT

Introduction: Italian and international outbreaks of highly pathogenic avian influenza (HPAI), particularly in densely populated poultry areas (DPPAs), have increased over the past few decades. These emerging risks, which endanger both human and animal health and the entire poultry industry, can be effectively limited by biosecurity measures implemented at human-animal food chain interfaces. Some problems, however, persist in the application of these measures on the part of poultry farmers, prompting the need to explore those aspects and causes that limit their implementation.

Material and methods: A qualitative approach was selected for the study and a semi-structured interview technique was applied to collect data among turkey farmers ($n = 29$) working in the north-east of Italy. The aim of this technique was to gather data on farms in order to understand the biosecurity practices adopted and the reasons for and impediments to farmer implementation, or lack thereof. This article presents and discusses the main data collected.

Results: The study revealed that farmers were familiar with the biosecurity measures necessary to contain avian influenza (AI) and other poultry diseases; personal disinfection and animal isolation practices were particularly prominent. Based on the reported procedures, managerial, economic, and psychosocial factors were among the barriers behind the failure to implement biosecurity measures. These obstacles were variously intertwined and associated with the different action settings. In particular management factors, such as lack of time to apply the rules and difficulties contingent on the farm's structural characteristics, mediate the application of biosecurity measures. In terms of communication channels, the company, particularly its technicians, proved to be the primary source of information for farmers in case of emergencies, as well as the primary source of information on the application of biosecurity measures. However, other sources of information were indicated, such as word of mouth among farmers or other non-institutional figures (relatives and acquaintances).

Conclusions: What emerged, was the need to improve not only the biosecurity management skills, but also to implement forms of cooperation among the various key stakeholders in the poultry sector. The information presented in this pilot study needs to be discussed among competent authorities, public and company veterinarians, company technicians, and farmers. Furthermore, this information will help in participatory co-planning of risk prevention and communication strategies to implement a long-term, sustainable, effective approach to address future epidemic emergencies.

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1. Introduction

In Italy, the avian influenza (AI) recent emergency started in the 1990 s (Adlhoch et al., 2022; Capua et al., 2002; Chatziprodromidou et al., 2018; Fusaro et al., 2017; Mulatti et al., 2018), and in the last years the number of outbreaks greatly increased, involving in particular the most highly densely populated poultry areas (DPPAs). These regions are also characterized by the presence of wetlands along migratory routes and the wintering sites of numerous species of wild birds, known as reservoirs of influenza viruses. The introduction of viruses into industrial poultry farms can cause severe epidemics, with massive economic impact on the poultry sector, and posing a threat to animal and human health (Abbate et al., 2006; Adlhoch et al., 2022; Chatziprodromidou et al., 2018; Lambrou et al., 2020).

A National Surveillance Plan For Avian Influence is designed and implemented by the Italian Ministry of Health on an annual basis to reduce the risk of HPAI uncontrolled spread (Ministero della Salute, 2023). Although these measures are strictly applied during epidemics, this is not always the case during inter-epidemic periods, when inadequate attention is paid to the risk of HPAI. This suggests the need to implement integrated risk management through involvement and active participation of the various stakeholders, in particular the farmers. Emergency plans need to go beyond a coercive top-down system (Alders et al., 2020; Barnett et al., 2020) and incorporate more participatory epidemiology, designed to intercept needs and coordinate the different skills of the actors involved, with a view to developing a transdisciplinary approach to disease control (Hinchliffe, 2015).

Social research techniques have proven to be a useful tool for implementing participation in the veterinary epidemiology field (de Oliveira Sidinei et al., 2021). Failure to incorporate this local knowledge and experience can lead to misplaced interventions that are unsustainable in the long term (Alders et al., 2020; Barnett et al., 2020). Numerous studies have explored farmers' behaviour, knowledge, and opinions on biosecurity measures, using social research methods and techniques and providing epidemiologists with new insights (de Oliveira Sidinei et al., 2021; Delpont et al., 2021; Holloway, 2019; Lambrou et al., 2020; McPake et al., 2022; Naylor et al., 2018; Paul et al., 2015; Porter, 2019; Scott et al., 2018). The available scientific literature highlights that compliance with farm biosecurity measures can vary according to personality traits, sense of responsibility, personal work experience, and sociodemographic factors (Bánkuti et al., 2020; Holloway, 2019; Hyland et al., 2018; Racicot et al., 2012; Simon-Grifé et al., 2013). Also, factors such as lack of knowledge about diseases, the economic cost of implementing these measures, workload, and aversion to imposed procedures, are suggested as possible barriers to their application (Lambrou et al., 2020; Mankad, 2016; Moskalenko et al., 2022; Renault et al., 2018; Ritter et al., 2017; Sarrazin et al., 2014; Scott et al., 2018; Simon-Grifé et al., 2013). Other factors affecting farmers' decision-making processes include socioeconomic characteristics, family relationships, and the farming company's organizational structure (Bánkuti et al., 2020; Indrawan and Daryanto, 2020; Noremark et al., 2010; Sayers et al., 2013; Shortall et al., 2018). These premises point to an emerging need to investigate the difficulties of applying biosecurity measures to prevent HPAI outbreaks in the poultry sector. In the current work we reports the main results of social epidemiology research, involving poultry farmers operating in North-East Italy, with the aim to identify the main motivations and obstacles to applying biosecurity measures and to plan appropriate prevention and communication risk strategies on AI.

2. Materials and Methods

2.1. Research approach

In line with the leading participatory epidemiology studies (Ebata et al., 2020), a qualitative social research approach was adopted in the study presented in this paper. Twenty-nine semi-structured interviews (Bryman, 2016) with poultry farmers were conducted to explore the daily practices adopted in farms, the rationale behind them, and any aspects that could lead to increased risks of pathogen introduction. A social researcher guided the interviewees through a standard trace outline, while leaving ample space for spontaneous themes and reflections during conversations. The interview structure included four thematic sections: characteristics of the respondents, meaning of biosecurity practices on the farm and their application, factors limiting the application of biosecurity standards on the farm, and communication tools. The interview guide was developed following suggestions from the sociological literature (Becker, 2008; Corbetta, 2003).

A single social researcher conducted the face to face interviews between March and April 2016. The interviews about one and a half hours and were held directly at the farmers' homes or on the farm to allow contextualization of the narrative data. After obtaining permission from the integrated poultry company to contact the farmers, the selected participants gave their informed consent to take part in the study and to comply with the General Data Protection Regulation (EU) 2016/679 (European Parliament, 2016), a privacy agreement was request to all participants involved in the research activities and a paper document was signed by farmers before the interview began.

The conversations were audio recorded and transcribed verbatim; the information obtained was manually coded by an interdisciplinary working group through thematic analysis (Bryman, 2016) and reorganized through groupings of specific practices, with the aim to identify specific topics and related sub-topics. To ensure better understanding, the data were summarized in thematic tables in which the text was organized logically around the objectives of the analysis. The tables include the frequencies of the data. The sum of the frequencies can exceed the number of interviewees since they indicate several aspects referring to a particular practice or identified category. Extracts of the dialogues were reported anonymously to enhance data interpretation.

2.2. Participant recruitment

Meat turkey farmers located in Verona province were chosen for inclusion in the study. This area has a high density of meat turkey farms and has markedly been affected by AI epidemics over the past years (Marangon et al., 2004; Mulatti et al., 2010). To better collect data on the application of biosecurity measures and the communication methods used in farms, only farms belonging to a single integrated poultry company were included in the surveys. Approximately 90% of the Italian poultry production system is organized in a semi-vertical integration system, consisting of a few integrating companies which provide all the necessary support, such as birds, equipment, technical and medical assistance to a large number of integrated farmers (Ministero della Salute, 2023).

In total, 29 turkey farms were selected through a stratified random sampling method, based on the following criteria: a) herds with or without H9N2 virus infection history; c) presence or absence of external employees involved in company management; d) gender of the farmer; e) willingness to be involved in the interview.

3. Results

3.1. Characteristics of the farmers involved

The selected sample was mainly composed of male farmers (male = 22; female =7), with an average age of 50.4 years, working on a family-run farm. 14 out of 15 farmers have been affected by avian influenza in the past (Table 1).

3.2. The meaning of biosecurity: farmers' practices and experiences

The analysis of the interviews allowed to explore the meaning

Table 1
Characteristics of the meat turkey farmers involved in the interviews (n = 29).

Farmer ID	Gender	Age	Presence of employees	Family member employed	Previous exposure to AI
1	male	33	-	✓	-
2	female	45	✓	-	✓
3	female	38	-	✓	-
4	male	59	-	✓	✓
5	male	73	-	✓	✓
6	male	44	-	✓	-
7	male	42	-	✓	✓
8	male	63	-	✓	-
9	female	50	-	-	-
10	male	62	-	✓	✓
11	female	23	-	✓	✓
12	male	51	-	✓	✓
13	male	66	-	✓	✓
14	female	36	✓	✓	✓
15	male	45	-	✓	-
16	female	46	-	-	-
17	male	67	-	✓	-
18	male	67	-	✓	✓
19	male	60	-	-	-
20	male	44	-	✓	-
21	male	52	✓	-	-
22	male	39	✓	-	-
23	female	40	✓	✓	-
24	male	46	✓	-	-
25	male	57	✓	-	✓
26	male	76	✓	✓	✓
27	male	33	✓	✓	✓
28	male	44	✓	-	✓
29	male	63	✓	-	-

Table 2
Biosecurity measures perceived as relevant by meat turkey farmers (n = 25)[a].

Measures*	Implementation	Specific procedures
Disinfection (n = 47)	-Disinfection of premises, equipment and vehicles (n = 22)	-Disinfection of vehicles (n = 7) -Rodent control operations (n = 6) -Cleaning (generic) (n = 5) -Disinfection of concrete operational areas (n = 2) -Prohibiting machinery from leaving the farm (n = 1) -Antimosquito treatment (n = 1)
	-Personal disinfection (n = 15)	-Disinfection of boots and hands before entry (n = 9) -Changing of footwear and wearing of specific clothing (n = 5) -Creation of filter areas (n = 1)
	-Disinfection (generic) (n = 6)	
	-Disinfection of external staff (n = 4)	-Controlling access of external staff (n = 3) -Making staff wear appropriate clothing (n = 1)
Isolation (n = 20)	-Restriction of access (n = 8)	-Reduction of visits (n = 8)
	-Preventing the entry of animals (n = 7)	-Adoption of anti-bird nets (n = 6) -Prohibiting pets to be kept on the farm (n = 1)
	-Monitoring and recording access (n = 3)	-Registration of visitors entering the farm (n = 3)
	-Mitigating virus entry (n = 1)	-Adoption of forced air systems (n = 1)
	-Fencing (n = 1)	-Fencing the farm perimeter (n = 1)
Removal (n = 1)	-Poultry manure/litter storage (n = 1)	-Construction of a manure storage pit (n = 1)
Other (n = 5)	-Creation of restricted areas (n = 1) -Farms divided by animal sex (n = 1) -They are all important (n = 3)	

[a] 4 farmers did not provide any information on this topic
* This column reports the overall frequencies of the coded information. If the number is higher than the total number of respondents (n = 29) it means that they provided more answers/information regarding the issue investigated

farmers attribute to the concept of biosecurity and their experiences with it in context. Among those providing definitions (n = 24), the prevailing meaning of biosecurity was the 'protection of animals from pathogens' (n = 19); followed by the 'farm's hygiene status' (n = 5), the ability to 'consciously organize own work' (n = 4), and 'worker safety' (n = 3).

Table 2 summarizes the farmers' opinions on the biosecurity measures they considered relevant in protecting farms from the introduction of pathogens.

Farmers believed that properly applied biosecurity standards consisted mainly of disinfection (n = 47) of premises, spaces, people, equipment, and vehicles, and isolating the animals from the outside (n = 20). Among the reasons given, farmers pointed out that these measures mainly served to protect animals, make them stronger and, therefore, more capable of counteracting disease onset; in addition, these precautions helped produce good quality meat:

Nowadays, you can't do this job without observing specific rules to protect the consumer if I make an animal on the counter that has taken more drugs than usual, but if I can get it to arrive with less drugs, it means that the meat is much more genuine. (ID 12).

Some farmers talked about the most significant transformations characterising poultry farming of late. They highlighted that the HPAI epidemic in the 2000 s constituted a critical watershed for regulatory and cultural change in terms of virus spread containment. The crisis significantly increased the problems associated with conducting this line of farming activity. The new avian threat put the sector into serious difficulty and some farmers reported their experiences with the outbreaks on their farms. Emotional aspects also emerged about the mass deaths of animals and the imposition of culling procedures during emergencies.

One morning in '99 […] I entered a shed with about 8000 turkeys that were ten days old: they were all dead, in a sea of white. And we knew that we had to kill all the turkeys in the other three tunnels but didn't have the courage […] So experiencing these emotions teaches you to take certain precautions. (ID 02).

During and after this crisis, which spread to high-density areas, biosecurity measures were intensified:

Before the Avian emergency, I worked with another company, where everything was fine. In 1998, we changed companies and it immediately asked us to adopt a more rigid system; they understood that it was the only way to avoid diseases. (ID 20).

Over the years, various measures have been put in place to contrast HPAI. Some organizational and technological measures, accompanied by innovations and genetic engineering, have allowed the sector to continue growing. New technologies (forced air, automatic temperature control, etc.) enable environmental parameters to be controlled, increasing animal welfare. However, in some cases, this has allowed animal density to increase further.

3.3. *Health problems are often exacerbated in a closed environment, such as a high density sheds. (ID 01)*

However, new trends are emerging that run counter to the unlimited growth tendencies of the intensive sector. Some farmers who had started working for the German market (ID 08, 17, 28, 29) reported having to reduce the number of turkeys and animal density to ensure better animal welfare:

And then came animal welfare, which is a very important issue. I am now producing for the German market and have had to make changes: fewer turkeys per square meter, so they are less stressed; there have to be toys to keep them occupied; everything has to be fenced. (ID 28).

Animal welfare also affects animal health and the quality of the final product:

Animal welfare and biosecurity are not identical, but must go hand in hand. You can't have one without the other. (*ID* 14).

3.4. Identification of biosecurity practices

The interviews enabled to identify the leading biosecurity practices implemented by farmers, specifically, behaviours relating to personal disinfection (Table 3); disinfection of vehicles and equipment (Table 4); procedures adopted for external staff (Table 5); and finally, methods of storing and disposing of manure/litter poultry (Table 6).

Regarding the use of dedicated, disinfected boots (n = 29), 7 out of 29 farmers reported that each shed had its own set of boots or footwear, while the majority of farmers (22 out of 29) used one pair of boots only. Of these, 20 disinfected them only at the entrance, while only 2 disinfected them at both the entrance and exit. Hence adoption of the best practice of equipping each shed with several pairs of boots appeared to be limited, and their disinfection at both entry and exit was not widespread. Regarding the correct implementation of this practice, each farmer had different opinions on, for example, contact time of the disinfectant, the frequency of its replacement, and the use of brushes to remove soiled residues from soles. Furthermore, some company technicians enforced the installation of the so-called "Danish entry system", a physical barrier that does not allow personnel direct access to the area where the animals are located without first wearing dedicated footwear, which some consider to be better than just boot disinfection (ID 7); others find it hinders work a little (ID 6).

As for protective clothing practices (overalls, gloves, headphones) (n = 29), 26 out of 29 farmers reported always wearing specific clothing before entering the sheds. However, 25 used the same type of clothing without differentiating between animal contact areas and outside areas. 21 out of 29 farmers used the filter area to wear appropriate clothing, while 4 did not use it, with clothing being worn at home since it was adjacent to the sheds. In some cases, the filter areas were used as offices and for other practices such as document and equipment storage. Their location and conformation largely depended on the structure of the farms themselves and on the period in which they had been built: the buildings in most of the visited farms dated back 30 years. These secure access areas were therefore influenced by structural and logistic

Table 3
The personal disinfection practices among meat turkey flocks under study (n = 29).

Practices	Implementation	Specific procedures	Reasons
Disinfection of boots (n = 29)	One pair of boots (n = 22)	-Disinfected only at the entrance (n = 20) -Disinfected at both entrance and exit (n = 2)	-To isolate boots that have come into contact with the external source of contamination
	One pair of boots per shed (n = 7)	-Each shed is specifically equipped with boots and a disinfectant tray (n = 6) -Only one pair of boots is used but different shoe covers, immersed in disinfectant, are adopted for each shed (n = 1)	-The most appropriate measure and the one most in line with company requirements
Wear protective clothing (n = 29)	Use of specific clothing (n = 26)	-The same clothing for indoor and outdoor areas (n = 25) -Differentiated clothing for outdoor and indoor areas (n = 1)	-The outdoor area as a whole is considered a source of contamination: a distinction is made between the "dangerous" exterior and the interior "to be protected"
	Occasional use of specific clothing (n = 3)	-Overalls and gloves at specific times in the production cycle (n = 3)	-High temperatures inside the shed, overalls, and gloves make work difficult -Chicks only move inside the chick brooder and the farmer is unlikely to be touched by feathers, possibly soiled with feces
Presence of filter areas (n = 25)	Use of filter areas (n = 21)	- Filter areas are separate from the entrance to the sheds (n = 14) -Filter areas are adjacent to the shed entrance door (n = 6) -Filter areas are adjacent to the sheds (n = 1)	-A protected space guarantees enhanced shed insulation
	No filter areas are used (n = 4)	-Clothing is worn at home (n = 4)	-The farmer's house is adjacent to the sheds

Table 4
Disinfection practices for vehicles, operating equipment and machinery (n = 20)[a].

Means and tools	Procedure	Reasons
Feed and gas trucks (n = 20)	-Manual wheel disinfection (n = 11)	-Pipes tend to freeze in winter -Drivers dislike the smell of disinfectant -The disinfectant is sticky and remains on the surfaces -There is no room to install the disinfection arch -The disinfection arch is expensive
	-Manual disinfection of the sides of vehicles and the interior parking areas (n = 5)	-The exact reasons are listed above -Manual disinfection helps disinfect concrete operational areas, believed to be infected by pathogen-carrying wheels
	-Presence of the disinfection arch (n = 4)	-Easy to use, especially effective after changing the disinfectant (the previous one tended to clog the nozzles)
Tractors with a milling machine (for grinding poultry manure/litter) and wheelbarrows (for carrying carcasses) (n = 19)	-Wheel disinfection (n = 14)	-The wheels could introduce the feces of wild contaminated animals
	-Limiting the use of vehicles to specific farming activities (n = 3)	-In the case of dedicated vehicles (i.e. small tractors and wagons), this avoids carrying contaminated feces from one shed to another
	-Both wheel disinfection and limitations on vehicle use (n = 2)	-Using dedicated machinery (tractors) avoids carrying contaminated feces from one shed to another. The wheels are disinfected in any case
Machinery (generic) (n = 1)	-Disinfection, especially the wheels (n = 1)	-The wheels could introduce contaminated feces; for this reason, the machinery is kept outside the shed

[a] 9 farmers did not provide any information on this topic

Table 5
Biosecurity procedures for external staff in turkey meat farms.

External staff	Procedures (what the farmer provides)	Reasons
Procedures for veterinarians and technical personnel (n = 29)		
Veterinarians and company technicians (n = 29)	-Registration at the entrance; provision of overalls and boots; performance of personal disinfection practices (n = 19) -No information (n = 10)	-They are aware of the risks
Public veterinarians (n = 25)	-They come to the farm in coveralls and put shoe covers on before entering (n = 11) -Lack of information (n = 14)	
External workers (n = 1)	-Registration at the entrance; provision of coveralls and boots (n = 1)	-External workers are deemed to present a danger
Procedures for other personnel (n = 14)[a]		
Staff from cooperatives (n = 8)	-Provision of coveralls and boots; they must arrive 'clean' (n = 6) -No provision of clothing. They must arrive 'clean', bringing disinfected coveralls and boots before entering the shed (n = 1) -Not specified (n = 1)	-A relationship based on loyalty means farmers know "who they are" and "how they behave"; this generates reassurance in relation to health risk
Family members, employees, acquaintances (n = 6)	-No provision of clothing. They must arrive 'clean', bringing coveralls and boots with them, which are disinfected before entering the shed (n = 2) -Provision of coveralls and boots and the requirement to arrive "clean." (n = 1) -Not specified (n = 3)	-They do not visit other farms and are thus considered safe

[a] 15 farmers did not provide any information on this topic

variables ranging from proximity to the farmer's home to shed location.

The interviews revealed that all farmers believed the source of contamination came from stepping on wild animal droppings, prompting the need to disinfect clothing and footwear. Furthermore, the outside area was considered to be the primary source of risk for any internal contamination, underlining the distinction by which the exterior was perceived as 'dangerous' and the interior as a place 'to be protected'.

Another disinfection practice involved the vehicles, machinery, and equipment used on the farm (Table 4).

Like the boots indicated in the previous table, respondents selected vehicles as a source of contamination. In particular, the vehicles considered the most critical were 'feed and gas trucks' (n = 20), because they came from the outside and can bring in contaminated feces through their wheels. Farmers also paid attention to 'tractors with a milling machine and wheelbarrows' (n = 19) because the wheels could introduce contaminated animal droppings from the outside and from one shed to another. Carcasses are taken out of the shed and loaded onto wheelbarrows. They are transported to cold storage bins, which are kept away from the birds and emptied only at the end of the production cycle. Equipment such as the disinfection arch for incoming trucks were used by only 4 out of 20 farmers; the others preferred manual devices, which above all allowed the vehicle wheels to be washed.

Various types of staff can come into contact with the farm: permanent employees, temporary employees, and family members. Other external figures include company veterinarians, company technicians, public veterinarians, staff from cooperative societies, external workers, and acquaintances. Farmers adopt different procedures to contain any risks arising from contact with people from the outside (Table 5).

Table 6
Procedures for removing poultry manure: storage and disposal (n = 27)[a].

Manure management	Procedures	Reasons
Storage and/or disposal with private company (n = 17)		
Disposal with private company (n = 10)	-It is taken away a few days after the end of the cycle and the truck does not need to be disinfected (n = 9) It is taken away a few days after the end of the cycle, but it is always preferable to disinfect the truck (n = 1)	-This is considered the safest and easiest system -Unwillingness to build a manure bin/pit (costs) -No ownership of agricultural land -Disposal-related risks are avoided by using other colleagues
Storage and disposal with a private company (n = 7)	-Storage in a manure bin before being used on the land: concrete plateau with/without sides and a nylon cover, built away from the shed (n = 4) -A part is stored in the sheds to dry before delivery to the company (no more than a week), and part is stored in specially rented fields (n = 1) -A part is immediately delivered to the disposal company (n = 1) -Storage in a manure bin before being used on the land: area on concrete rather than on earth with a nylon cover, built away from the sheds (n = 1)	-Own land is not sufficient for complete disposal -In some cases, it first needs to be stored because some disposal companies only accept it dry
Storage and disposal through agreed delivery to fellow farmers (n = 6)		
Storage and disposal through agreed delivery to fellow farmers (n = 5)	-Before being used, it must be stored in a manure bin/pit to allow fermentation (n = 3) -A part is immediately used on the land or delivered to fellow farmers (n = 1) -At the end of the cycle, part of the manure is picked up and taken to the fields; part is delivered to fellow farmers. If fellow farmers cannot remove it immediately, or it cannot be used directly in the area, it is stored in a manure bin/pit, built of concrete with sides and a nylon cover, located away from the sheds (n = 1)	-Own land is not sufficient for complete disposal
Disposal through delivery to fellow farmers (n = 1)	-It is removed straight after the end of the cycle (n = 1)	-Fellow farmer interested in removing it all
Private storage and disposal (n = 4)		
Independent storage and disposal by the farmer (n = 4)	-Disposal a few days after the end of the cycle -Part of the manure is left to ferment in the sheds (no more than a week) -Part of the manure is placed in a bin with sides and a nylon cover, located away from the sheds (n = 3) -It is removed a few days after loading (it is partially fermented in the sheds for no more than a week) and then placed in a dedicated structure: concrete plateau	-Ownership of agricultural land -It cannot be used immediately on the land because it must be fermented and dried before use; moreover, the time for using manure on the land does not always correspond to the end of the farm's production cycle

(continued on next page)

Table 6 (*continued*)

Manure management	Procedures	Reasons
	without sides but with a nylon cover, built away from the sheds (n = 1)	

[a] 2 farmers did not provide any information on this topic

The farmers provided information on targeted measures for company technicians and veterinarians (n = 29) and public veterinarians (n = 25). Fourteen farmers provided information on standards for other types of staff: staff from cooperatives (n = 8), family members, employees, and acquaintances (n = 6). The most widespread criterion for choosing external personnel from cooperatives was workforce loyalty: the same teams were always chosen and the farmers sought to create a personalized relationship with them based on trust. This did not prevent farmers from choosing to provide part of the necessary clothing, thus helping to increase the level of biosecurity. The prevailing criterion with regard to the employment of family members and acquaintances was to avoid personnel who were in contact with other farms. Farmers could sometimes lower the burden of providing everyone with the necessary clothing. This choice was also based on economic and logistic reasons (inability to store all the material).

External personnel were involved in only certain stages of production, for example, treatment or immunisation teams, and any maintenance workers or feed truck drivers entered the safe area only. Not all farmers provided information on external personnel, which should instead always be recorded in accordance with legislation. The interviews revealed that farmers tended to avoid using external personnel as much as possible, perceiving them as a means of potentially introducing pathogens to the farm.

If a feeding trough breaks, you have to call someone from the outside, and it is challenging to check if they have animals at home or not [...]. Since we need electricians or other utility workers who don't know the risks, you often have to yell, 'Stop: don't come in with your shoes on!' (ID 29).

Since both public and company veterinarians are aware of the risks, farmers reported finding no difficulty in getting them to adhere to biosecurity procedures. Farmers reported that public veterinarians visited the farms only seldom, mainly on three specific occasions: audits relating to compliance with biosecurity rules; sampling in the event of AI outbreaks, and sampling for the health certificates needed to authorize animal loading and subsequent slaughter. On all these occasions, farmers reported using the same protective measures (coveralls and white coats, boots, or overshoes) provided also for company technicians and veterinarians. The analysis also revealed problems being experienced with the storage and disposal of poultry manure/litter (Table 6), which is a residue composed of poultry excrement and can constitute an environmental and biosecurity hazard when not disposed of correctly. At the end of each cycle, corresponding to the established sanitary vacuum periods, the sheds must be emptied, cleaned, and disinfected. Poultry manure/litter must be treated adequately and transformed through differentiated procedures before being reused as a fertilizer. Several factors can affect these storage and disposal procedures, as the presence or absence of own agricultural land, which may not permit complete disposal of the manure; possession of storage facilities for conserving droppings; compliance with the rules of each disposal company. There are also economic reasons, for example, failure to build a manure storage pit, or biosecurity reasons for choosing one system rather than another, in addition to the presence of different local regulations.

Three main disposal and storage methods were reported: storage and disposal through an external company (n = 17), delivery to a fellow farmer owing agricultural land (n = 6), and direct management by the farmer (n = 4). There could also be a combination of these methods, as described by one farmer:

The poultry manure is kept inside for 7/8 days to allow it to ferment. Afterwards, it is taken away, and I have a manure store with covered walls. When possible, what we can't dispose of on our own land, we give to another farmer or company [...] I have seen some farmers take it away immediately, when it is still dusty, even on a windy day. Doing it elsewhere can be detrimental. It can have a boomerang effect because it damages whoever does it. (ID 26).

Among the various procedures chosen by farmers, delivering the manure to fellow farmers was not a common disposal method (n = 6).

A farmer once came to get the manure. And I told him, 'Listen, you should cover it'; 'But I only have to cross the street', he responded, which was true. But he was stopped, and they gave us a fine. [...] Since then, I decided to give it to a private company; they come with the truck and cover it. (ID 08).

Several concerns were thus voiced in relation to disposal through fellow farmers, including incurring fines in case it is not transported in accordance with legislation. In addition, during the interviews, some farmers complained about the performance of fellow farmers, who do not always comply with the sanitary vacuum periods and the rules governing manure transport.

It is a question of loyalty; for example, they should load the manure at the end of the cycle and transport it in an open truck. If there is nothing going around at the time, it will be ok! But if there is an emergency warning, it becomes dangerous. I've had experience of these things. (ID 26).

3.5. Factors limiting the application of biosecurity standards on the farm

Farmers were asked for their opinion on aspects limiting the adoption of biosecurity practices. Data revealed that the discrepancy between legislation and daily routines could depend on multiple factors (Table 7), including managerial (working times, contextual constraints, relationships with staff) (n = 30), psychosocial (personal characteristics, generational differences, resistance to change, fatalistic attitude) (n = 19), economic (investments and incentives relating to perceived effectiveness) (n = 6), and cognitive factors (attribution of uncertain randomness) (n = 4), which can be variously intertwined within different contexts of action.

Farmers considered management factors (n = 30) associated with the organization of work practice to be most able to trigger non-compliance. This macro-category groups together aspects more specifically linked to work organization and relations with employees or external staff (e responsible for supplying feed, gas, etc.); procedures for carrying out work tasks and the time taken to complete them; contextual constraints, relating to the farm's spatial location, or structural limitations.

The 'working time' category refers to slowdowns resulting from the

Table 7
Factors underlying failure to comply with biosecurity standards.

Aspects*	Farmers reporting aspects of own breeding practice (n = 16)[a]	Farmers reporting aspects of breeding practice in general (n = 21)[b]
Managerial (n = 30)	Working times (n = 9) Contextual constraints (n = 6) Working with staff (n = 2) Requests to adapt equipment (n = 1)	Working times (n = 7) Contextual constraints (n = 2) Multiple entrepreneurial activities (n = 2) Relations with staff (n = 1)
Psychosocial (n = 19)	Generational differences (n = 1)	Personal characteristics (n = 9) Generational differences (n = 7) Resistance to change (n = 1) Fatalistic attitude (n = 1)
Economic (n = 6)	Investments and incentives (n = 2)	Investments and incentives (n = 2)
Cognitive (n = 2)	Uncertain causality (n = 1)	Uncertain causality (n = 1)

* This column reports the overall frequencies of the coded information. If the number is higher than the total number of respondents (n = 29) it means that they provided more answers/information regarding the issue investigated
[a] 13 farmers did not provide any information on this topic
[b] 8 farmers did not provide any information on this topic

application of work practice rules:

In the past, nothing got disinfected, and everything was done quickly. Having to stop and get another tractor to connect the pump, disinfect everything and be careful requires a lot of time. And all of a sudden it's evening (ID 20).

The extra time taken to complete tasks is not, however, significant. The farmers themselves confirmed that the rules take a 'matter of a few minutes' to apply. It does instead seem to carry the psychological cost associated with any interruption to 'business as usual' caused by extraordinary events requiring prompt intervention (e.g., a pipe bursting in a shed, sick animals, etc.), in which case rules become 'troublesome' and challenging to implement (especially when they involve disinfection).

You end up going backwards and forwards from the shed 100 times. You can't do it every time, I'll be honest with you. it's tough to apply that stuff 100% of the time. We do what we can. (ID 15).

The 'working with the staff' factor, identified in interviews with farmers with employees, highlighted the challenges of conveying the content and meaning of biosecurity regulations to staff who are present on the farm less than the owners:

We are the farm owners and obviously use a different logic from employees. It is normal, in the evening before going to bed, for us to go and take a look at the animals and we are willing to work all night long if necessary, but at 5.30 pm the workers go home without worries: which is, as it should be. (ID 05).

The degree of involvement of the owner is central to employee training. This also depends greatly on his/her level of participation in this activity. Difficulties also arise when family members are involved, particularly in the case of elderly staff:

My dad is even more difficult than my employees. My dad doesn't believe these things are needed. [...] I saw my dad was the first to avoid doing things and in the end, I gave up. (ID 23).

The interviews revealed that rather than involving external staff, farmers preferred to involve family members, acquaintances, and trusted people. Farmers considered the use of external staff to be a risky practice. The most common criterion was to always choose the same external personnel from cooperatives or intermediation agencies. The owners of several farms tended above all to use external staff, while smaller owners, where there were family members, tended to do everything themselves:

Maybe the animals get less sick with us because we have a family business. Instead, those with lots of turkeys had to bring in external staff, who had also been on other farms. (ID 15).

In addition to family management, farmers also mentioned the size of the farms:

Before, maybe because we had six farms and we went around more, there seemed to be more diseases. Now I have fewer sheds and less animals. Perhaps it was because it used to be the area with the most farms, there are fewer farms here now, and so I'm not as badly affected by diseases because when you are in a high density area, the air carries the diseases around. If you have eight sheds, you can't do what I do (referring to the attention and treatments given to animals). (ID 09).

Farmers also attributed the introduction of the disease into farms to another set of factors, namely 'contextual constraints', i.e., elements associated with the characteristics of the farm location or the physical-structural characteristics of the sheds. The farm itself, especially in relation to existing buildings, needs to be adapted to growing safety demands.

I pay a lot more attention to the shed near the highway because it only takes a truckload of turkeys or a chicken infected with something to go past and move some biological material around; and trucks bring a lot of air. I have noticed that certain diseases always start from that shed. (ID 26).

Some farmers referred to a series of risk factors describing a complex ecology of relationships: high density, genetic variability among animals, proximity to highways or opportunist free rider farmers, involvement of external staff, wandering poultry (geese, hens), the presence of rats, coypus and small animals, birds released for hunting, or the proximity of pigeon, rabbit and laying-hen farms. Several farmers also mentioned air and wind as possible causes of pathogen spread (ID 01, 05, 08, 09, 10, 11, 12, 15, 24, 25, 26, 27, 28). This set of factors questions the extent to which certain constraints can nullify the effect of the adopted biosecurity measures. Economic factors also limit the adoption of biosecurity measures and refer to any financial burden resulting from proper application of biosecurity standards. The interviewed farmers reported making numerous investments, some of which proved sustainable in the long term, for which they recognized the benefits, despite the initial outlay:

Of course, adopting them meant making sacrifices, and I, too, lost some money, but there were benefits because it made all the companies get a bit in line; before there was chaos. (ID 12).

In addition to the requests made by the company technicians, farmers also reported economic penalties for failure to meet specific requirements, for which they were fully liable:

The company acknowledges when the animals don't get sick and when we use fewer medicines. In the end, they count the expenses, and the company gives you less if your medication costs were too high. (ID 04).

Older farmers claimed that before AI, the work was more straightforward; costs were now perceived to be rising and earnings to be decreasing. Some farmers were even considering closing the farm:

Before, you could afford many more things. If, for example, turkeys reached 13 kg, you made money anyway; if 20% died, you still made money. Now, if you had to load them at 13 kg or 20% die, you won't earn anything. (ID 19).

The 'psychosocial' category is a set of factors, including attitudes or behaviors, referring to self-attributed characteristics, particularly attitudes towards any transformations required by application of the rules. These characteristics are specifically related to working methods or broader attitudes, which go beyond the specific farm setting and concern instead the perception of risk or the possibility of controlling it.

The disinfectant should be changed once a day, they told me. Sometimes, out of laziness, you don't bother to do it; then you disinfect your shoes and the manure sticks to your soles, and maybe you step on some wild bird feces. The best thing would be to have a pair of shoes for each shed, and that is what I want to get to. I have to get used to it. remember to buy them, but then you have to explain that to your employees too. (ID 22).

A 'generational issue' was reported exclusively by farmers under the age of 55. Some young farmers argued that applying biosafety standards to some practices could be challenging for older farmers whose career started and was consolidated at a time when health culture was less relevant.

I try to explain things to him [my dad] and to motivate him, to let him know it's in my interest too, and get him to adapt. But I only have to turn my back for a moment and maybe he doesn't [...]. Now, with everything that has happened, we need to change. (ID 07).

Doubts were also expressed regarding the effectiveness of the biosecurity standards. Notably, these doubts did not amount to disrespect for the legislation, but revealed uncertainty about its ability to fully contain the entry and spread of pathogens. Despite complying with the rules, some farmers had a fatalistic attitude, stemming from their awareness of the diversity of conditions that allow the virus to be introduced and humans inability to control them.

It's almost impossible to achieve perfection in these procedures; theoretically, isolating animals from a potential source of contagion works. In practice, there are many variables. (ID 01).

The farmer has to deal with the actual conditions under which this legislation applies.

We should stay inside the bleach, but obviously, it is impossible. We carry diseases not only on our feet but also on our nose, eyelashes, and hair. So you do what you can. (ID 29).

Farmers pay greater attention in times of emergency compared to interepidemic periods, as some of them reported:

When everything goes well and there are no emergencies, we tend to leave *out certain practices. Sometimes we disinfect the truck quickly hoping there won't be any problems. Instead, we must be committed to disinfecting properly at all times because while nothing may happen today, we don't know what tomorrow will bring. (ID 19).*

[During emergency periods] the alert level is also too high, there is a risk of exceeding the measures and paying much more attention: for example, disinfection is more careful and more constant and you try not to neglect any detail. The farm is completely closed and only people in charge and feed trucks can enter (ID 25).

Finally, the fourth macro-category, i.e., 'cognitive,' refers to processes involved in knowledge of the outside world, specifically to identifying the uncertain causal link between the application of biosecurity procedures and their effectiveness.

Unfortunately, we are forced to fight invisible enemies. I makes me sick to think about it: when you can't see the enemy, you never know whether you have done a good job or not, or if you've done it well enough. What more can you do? You even try going inside less. (ID 20).

According to some farmers, 'it's not clear why you need to adopt all these rules', emphasizing the uncertainty of the causal relationship between the non-presence of the disease and the adoption of biosecurity measures:

I did everything I was told to do, and I still got salmonella. I have yet to understand how I got it; no one has been able to explain it to me. (ID 04).

Farmers were continuously in search of evidence and cause to determine the origin of the infection in their sheds, make improvements and avoid significant losses.

4. Communication to farmers

During the interviews two aspects of communication: farmers' communication channels during emergencies (Table 8) and their main sources of information on the application of biosecurity measures (Table 11). As shown in Table 9, the main communication channel during emergencies was the Company (n = 35) and technicians were the main vehicle (n = 20).

In addition to the alert statement, the content of the messages disseminated by the company was related to the biosecurity standards to be applied to reduce infections and safeguard livestock. Another relevant communication channel was the National Health Service veterinarian working for the Local Authority (n = 9), who usually provided information on the presence of a local outbreak. This type of information also came from neighbour farmers (n = 7) by word of mouth, and communication among fellow farmers was sometimes faster than official communications provided by the company, as reported below.

Once, a technician came and told me that about an outbreak of avian influenza. But I already knew because a farmer had already told me. (ID 08).

Table 9 presents farmers' evaluations of the various forms of communication. The findings highlight the prevalence of positive assessments of the communication messages sent by the company, the Local Authorities of the National Health Service, and other farmers working in the community.

As regards negative assessments of company communications (ID 06, 08 20, 26), several respondents reported that the company had a tendency to minimize the actual threat level:.

That is, they say as little as possible to avoid causing too much chaos. They tell us there are diseases in our area, but I would like to know more. (ID 06).

The information provided on AI is timely but not always complete, as in the case of other diseases:

When there is a case of avian flu, the company's technician sends us emails with everything that needs to be done to avoid contagion. However, I wasn't told anything about other diseases present in the area, such as mycoplasma. (ID 20).

It would be helpful to have more information on the epidemiological situation in the area. I apply all the biosecurity measures but I should be informed all the same if the farm opposite has infected cases. If an animal gets sick on my farm, I notify all my fellow farmers, but I don't always get this information from the company. (ID 08).

The company (n = 17) was the primary source of communications on issues of biosecurity not only during emergencies but whenever farmers needed any clarification or technical information.

Only 4 of the 19 respondents who indicated their main source of biosecurity information were aged over 50, and only one of them independently sought information through personal reading. The predominance of farmers under 50 pointed to generational differences in the assembly of knowledge about biosecurity standards. Older farmers reported expertise based on experience or preferred to establish personal relationship channels with technicians.

One 60-year-old farmer who managed his farm alone stated:

But having 16 years of experience, I know things too. For example, why

Table 8
Farmers' communication channels during emergencies (n = 28)[a].

Source*	Specifications	Tools	Type of information
Company (n = 35)	-Technicians (n = 20) -Company staff (generic) (n = 8) -Veterinarians (n = 7)	-Mail and phone call (n = 10) - Rapid messaging (i.e. SMS) (n = 1)	-Application of biosecurity measures (n = 15) -Increase in the security level (n = 5) -Awareness-building about biosafety (n = 1) -Outbreak location (n = 1) -Not specified (n = 13)
National Health Service - Local Authority - (n = 11)	-Veterinarians (n = 9)	-Direct relationships (n = 2) -Website (n = 2) -Not specified (n = 7)	-Presence of outbreak (n = 4) -Need to deny access (n = 1) -Not specified (n = 6)
Other (n = 13)	-Farmers (n = 7)	-Word of mouth (n = 7)	-Presence of outbreak (n = 7)
	-Institutional experts (n = 3)	-Website (n = 3)	-Presence of outbreak (n = 3)
	-Relatives (farmers) (n = 1)	-Word of mouth (n = 1)	-Rules to avoid contagion (n = 1)
	-Carcass disposal workers (n = 1)	-Direct relationships (n = 1)	-Presence of outbreak (n = 1)
	-CREV[b] (n = 1)	-Web (n = 1)	-Presence of outbreak (n = 1)

[b] Regional Center for Veterinary Epidemiology
[a] 1 farmer did not provide any information on this topic
* This column reports the overall frequencies of the coded information. If the number is higher than the total number of respondents (n = 29) it means that they provided more answers/information regarding the issue investigated

Table 9
Assessment of communication sources during emergencies (n = 28)[a].

Source	Positive assessments	Negative assessments
Company (n = 28)	Excellent / Effective (n = 5) Adequate (n = 13) Timely (n = 5)	Lack of information on AI (n = 2) Lack of information on other diseases (n = 2) Communication not prompt enough (n = 1)
National Health Service – Local Authority (n = 5)	Excellent / Effective (n = 1) Adequate (but random) (n = 2)	Lack of information on AI (n = 2)
Other farmers (n = 6)	Adequate (but random) (n = 5)	Not always reliable (n = 1)

[a] 1 farmer did not provide any information on this topic

Table 10
Farmers' main source of information on biosecurity (n = 19)[a].

Source	Specifications	Channel
Company (N = 17)	Technician (n = 8)	Direct relationship (n = 6)
	Veterinarian (n = 6)	Direct relationship (n = 4)
	Company (n = 3)	Courses, conferences, meetings (n = 3)
National Health Service – Local Authority (n = 1)	Veterinarian (n = 1)	Direct relationship (n = 1)
Other Sources		
Relatives (n = 1)	Close relatives (n = 1)	Direct relationship (=1)
Web (n = 2)	Websites (n = 2)	
Books / magazines (n = 1)		
Farmers (n = 1)		Direct relationship (n = 1)

[a] 10 farmers did not provide any information on this topic

can't you do cycles? Before, you could get up to 10 kg with female turkeys; now, you get 8 kg. I told the vet it depends on the feed, as my mother told me. But they [the company vets] are always right, and you always have to agree. They say 'it doesn't depend on the feed'; 'it depends on the species', 'it might even depend on genetics', but in my opinion, it depends on the feed. (ID 19).

Sometimes the information given by veterinarians contradicts the farmers' opinion. Some farmers felt expert knowledge was anchored to normative abstractness, and therefore far removed from the specificity of the settings.

Yes, in the sense that veterinarians are veterinarians, and we are farmers. We are a permanent part of breeding; we see the animals and perhaps understand how they are before the veterinarians do. Veterinarians always look at the rules on animal welfare etc. *Still, they don't provide all the answers because, for example, the humidity. it's raining outside, and we can't do much more, and they come here and tell us, "eh, the humidity is a bit higher than the standard" (ID 11).*

5. Discussion

5.1. Meaning, practice and experience of biosecurity

The data reported in the manuscript allowed to collect information on farmers' biosecurity practices perception. Findings show that these practices are applied by farmers primarily to the protection of animals rather than humans, in keeping with the scientific literature on the subject (Brennan and Christley, 2013; Sarrazin et al., 2014; Scott et al., 2018; Zaltron et al., 2021). The scientific literature shows that the AI virus is a significant zoonotic risk, prompting the need to adopt personal self-protection behaviours (Abbate et al., 2006; Adlhoch et al., 2022; MacMahon et al., 2008). These aspects warrant further investigation in future studies.

The interviewed farmers attributed efficacy and importance to the requisite biosecurity measures; belonging to an integrated farm system may have made them more aligned in their application (de Oliveira Sidinei et al., 2021). The scientific literature shows that adopting biosecurity practices, such as disinfection and isolation, helps reduce the risk of introducing pathogens into a farm (Guinat et al., 2020; Kwon et al., 2020; Yoo et al., 2022). The actions that gave farmers the feeling they had adequately applied biosecurity standards were above all the disinfection of environments, spaces, people, equipment, vehicles, and the adoption of forms of isolation, including the reduction of visits and animal isolation, again in keeping with other studies (Moskalenko et al., 2022). The adopted practices appear to be justified by farmers' belief that stepping on wild birds' faeces was the primary source of contamination. On the other hand, several responses to two of the rules helped to keep the spread of AI under control, namely the establishment of micro-areas and the division of farms by sex, enabling animals to be loaded and placed at the same time, resulting in a reduction in visits to the same farms.

While aware of legislation, the interviewed farmers highlighted shortcomings in its practical application: despite being one of the very first items at risk (Abbate et al., 2006; Lambrou et al., 2020), boots were not always properly disinfected at both the entrance and exit. Furthermore, appropriate clothing was not always used. In spite of some resistance, the integrated farm system is slowly moving towards implementation of increasingly stringent biosecurity measures. Greater attention was applied in times of emergency rather than in inter-epidemic periods, when alert levels are lowered, as declared by some farmers. Hence, a constant exchange of information must be maintained at these times. Numerous concerns were also expressed in relation to poultry manure management procedures which, given the risks it can create, need to be clearer and more systematic to better guide farmers' actions.

5.2. Factors limiting the correct application of regulations

The interviews revealed some social, cognitive, and psychological-behavioural factors that can lead to incorrect application of legislation and procedural guidelines. Management factors, such as lack of time to apply the rules and difficulties contingent on the farm's structural characteristics, mediate the application of biosecurity measures. In line with other studies (Noremark et al., 2010; Sayers et al., 2013; Simon-Grifé et al., 2013), findings suggested that greater attention was paid to biosecurity practices by owners of several sheds, exposed to more significant risks and more likely to seek external technical support. Interviewees from family-run farms preferred to do the work independently, but this could make it difficult to complete all planned and unplanned daily activities. However, some studies did not find a relationship between company size (number of animals) and greater compliance with staff biosecurity (Scott et al., 2018). Others showed that in larger companies it was not always easy to transmit the culture of risk to employees (Zaltron et al., 2021). Furthermore, a 'generational issue' emerged consistent with other studies (Hyland et al., 2018; Zaltron et al., 2021): those most likely to change their behaviour were younger farmers, who are most knowledgeable about AI (Abbate et al., 2006).

Psychological and cognitive factors played a key role in farmers' stories, including specific resistance to change in the presence of established work habits, lack of motivation, and, in some cases, a fatalistic attitude. These aspects were reported also in other studies in the field (Richens et al., 2018; Shortall et al., 2016). A fatalistic attitude refers to a situation in which expectations are frustrated by both discipline, which makes work demanding, and facts, such as epidemics, perceived to be inevitable, out-of-control phenomena. In line with other studies (Joffe et al., 2016), this fatalistic attitude mediates the relationship between risk awareness and risk prevention behaviours. Some farmers, in corroboration of other studies (Naylor et al., 2018), identified several external risk factors that point to a complex ecology of relationships. These factors were perceived to be beyond their control and to increase the incidence of AI. This was particularly evident in the management of farm porosity. These and other previously reported assumptions call into question the effectiveness of the adopted biosecurity practices and shed light on the indeterminacy of human-animal-environmental connections. Farmers may instead be distorting reasoning for self-justification purposes and to attribute responsibility to others or to external causes (Bandura, 1996), besides protecting their professional identity and reputation as a good farmer (Naylor et al., 2018).

Consistently with other studies (Zaltron et al., 2021), some farmers stressed that there was 'no clear need to adopt all these rules', emphasizing uncertainty about the causal relationship. The perceived lack of their efficiency, feasibility, and usefulness were the most frequently cited reasons for failure to implement biosecurity standards and procedures (Mankad, 2016; Renault et al., 2018; Ritter et al., 2017).

Farmers considered disinfection actions and isolation elements, such as anti-bird nets and visit reduction, to be essential measures for protecting farms from pathogens. At the same time, however, they had doubts about other aspects of legislation or company requirements. These included the proposal to have a tractor for each shed for manure milling purposes, the installation of forced air systems, a light-adjusting clock, water collection tanks for truck disinfection, and the construction of bathrooms and showers, to mention just a few. According to the literature, the perceived cost-effectiveness factor is central to the adoption of biosecurity practices and tools (Ritter et al., 2017), whose effectiveness is not immediately evident but appears to be more a question of probability and luck. Farmers must take full responsibility for implementing any investments required to comply with legislation; and while these requirements are understandable for veterinarians, they are instead frustrating and sometimes counterproductive for farmers (Shortall et al., 2018). Having too many regulations and excessive requests - or “the process of alienation from bureaucratic life” - becomes almost as inefficient as the condition of anomie or the absence of precise regulations. The ’optimal’ situation, which has been intensely debated in the sociological field, seems to lie on an indeterminate boundary between anomie and alienation (Acevedo, 2005). This prompts the need to reflect on the practical impact of rules and how new regulations imposed from above, in the absence of co-construction and involvement, can interact with the ecology of those that already exist.

Although biosecurity measures are necessary to protect animals in high-density areas (Guinat et al., 2020), this approach nevertheless proves problematic when viewed from a long-term perspective. In recent decades the number of poultry farms has increased ‘irrationally’ in the North-East of Italy (Ministero della Salute, 2023). In this growth-oriented perspective (Lambrou et al., 2020), it is difficult to completely isolate large numbers of animals from the outside environment. In any case, they need environmental influences (e.g., contact with the farmer) to grow appropriately and meet integrated farm company standards (Beck, 2009; Delpont et al., 2018). Moreover, consumers are increasingly sensitive to products’ ethical credentials (Brombin et al., 2019; Cavazza and Guidetti, 2020). The permeability of walls is a requirement for life to live and total closure, envisaged by the biosecurity paradigm as the ideal situation, remains a complicated process (Hinchliffe et al., 2013).

The sector is subject to continuous change and poultry farms are under increasing health pressure (Hamilton, 2018), including from new, more incisive biosecurity and animal welfare regulations. Biosecurity measures create more work-related difficulties for small than for large farms. This can foster a fatalistic, resigned attitude among farmers who, while adjusting, struggle to see the positive long-term economic effects of investments made to keep up with the times and growing epidemic risks. This paradigm discourages economies alternative to intensification, favouring more capitalized, biosecure ones (Hinchliffe and Ward, 2014; Noremark et al., 2010; Shortall et al., 2018), despite their being the most implicated in recent outbreaks due to their high density (Holloway, 2019; Lambrou et al., 2020; Wallace et al., 2015). This point was raised by several farmers with family-run farms. The biosecurity paradigm, albeit a necessary part of the current system in high-density areas, is part and parcel with the consumption and growth trends of global capitalism (Meadows, 2008; Wallace et al., 2015) and some farmers were well aware of this. To ensure the applicability of biosecurity measures, farmers need to understand the importance of the biosecurity paradigm in the current framework, highlighting the economic and social and health advantages it can generate if adopted collectively and in a coordinated manner (Sarrazin et al., 2014). Its application would not only make production systems more resilient and capable of adapting to the growing environmental and epidemic perturbations, but would also improve farm animal welfare and indirectly improve farmers’ job satisfaction (Brennan and Christley, 2013; Sarrazin et al., 2014).

5.3. The communication of outbreaks and biosecurity measures

In terms of knowledge building and information sharing, the farmers in our survey were able to draw on a plurality of information sources, in line with other studies (Abbate et al., 2006; Cui et al., 2019). The interviews revealed that the company was the main source of communications regarding outbreaks, primarily through technicians. In addition, many farmers maintained that emergencies were communicated effectively by both the company and the competent public authorities, providing warnings and indications on how to avoid contagion. Interestingly, information was disseminated by word of mouth among farmers and other non-institutional figures. While this suggests an entrepreneurial, participatory attitude, on the part of some farmers, to the reconstruction of individual or group information, it could also be construed to show limited trust in the sources and content of company communications, or simply to indicate the need to have access to different information channels.

Farmers need to strengthen their collaboration with both public and company veterinarians and in particular to activate forms of mutual learning. In the absence of cooperative behaviours based on trust, a top-down approach becomes inevitable (Alders et al., 2020; Carr and Wilkinson, 2005). However, this approach is not sustainable in the long-term (Barnett et al., 2020; Ebata et al., 2020; Hoon Chuah et al., 2018) and can generate resistance. Epidemics are complex, interconnected phenomena characterized by great uncertainty and require multidimensional, multilevel governance. The interviews also revealed tensions between farmers’ practical know-how based on experience and experts’ knowledge. Some farmers considered expert knowledge to be anchored to normative abstractness and to lose sight of the specificity of the context.

Livestock and poultry production has, in recent decades, become more and more specialized, vertically integrated, concentrated (Bryant et al., 2022), and increasingly oriented towards precision farming. This has changed the role of the farmer, who has lost much of his/her decision-making and management autonomy. In the absence of cooperation in its various forms, these role changes can generate resistance, particularly in older farmers. In this complex scenario, farmers are regarded as a "bottleneck" in that their agency is intensely constrained by the other components of the system. As a result, they find themselves mediating between the various requirements and their own interests. Moreover, they operate in a complex social and environmental landscape characterized by uncertainty. To achieve an integrated and interdisciplinary approach, inspired by the principles of One Health, it is necessary to think in terms of systems (Meadows, 2008) rather than move with reductionist logic that impoverishes epidemiology by focusing on the attribution of problems to individual elements only, in this case farmers (Wallace et al., 2015). To play an active, informed, conscious role, farmers must not only be trained but also involved, understood, and listened to (Moskalenko et al., 2022). Furthermore, in the years to come, veterinarians, as the primary interface between institutions, companies, and farmers, will be instrumental in accompanying systemic improvements in biosecurity and animal welfare standards. This can only come from good knowledge of the target audience and the system as a whole.

A participatory approach in veterinary epidemiology can help co-build new practices with various stakeholders and specific communities, making it a useful approach to the prevention and control of diseases (Alders et al., 2020; Ebata et al., 2020). The non-compliance with the rules revealed by this study warrants discussion among the competent authorities, public and company veterinarians, company technicians, and farmers. However, existing resources will also have to be discussed, including the good practices that individual farmers have devised to address local issues, which could become formally recognized practices validated by experts. Implementing legislation is an ongoing process of constant dialogue and exchange, incorporating the specificities (limits but also opportunities) of individual contexts.

6. Conclusions

Our research work provides an accurate description of the behaviour of turkey farmers in North- East Italy, the biosecurity practices they adopt, and the importance they attribute to these practices. The study also explored several social factors related to improper adoption of biosecurity procedures. We found managerial, psychosocial, economic, and cognitive factors to be among the barriers to implementation of biosecurity measures by farmers. Qualitative social research methods are a valuable method for collecting contextual narrative data and opening the field to participation and collaboration (Ebata et al., 2020). This approach allows to highlight limits and possibilities that could help workable design solutions and alternative solutions to problems, as well as to plan targeted risk communication strategies. Our findings, which are specific to a high-risk area, should be considered complementary to the results of other studies conducted using more classical epidemiological methods. A social science approach would provide paramount insights on how risks factors for disease infections and related mitigation measures are perceived by the principal stakeholders (i.e. farmers), and hence should be integrated in a broader risk-assessment framework. In fact, a holistic approach accounting for both the presence/absence of biosecurity measures and the more individual-based aspects on how those measures are actually applied, would massively benefit the decision-makers in defining optimised control activities and allocating resources. Nevertheless, this would imply that such participatory approach needs to be applied on a much larger set of poultry farms, ideally covering the whole poultry farm population in north-eastern Italy. This study is a starting point for designing new investigations locally and nationally through participatory epidemiology. Our pilot analysis approach can be applied to other geographical areas and production sectors, in the belief that it will produce similar results on which to build effective risk communication strategies.

6.1. Limits and developments

This study was focused only on farmers from a single integrated farm company however this company stands out among the main ones in the poultry sector in Italy. Further research activities can however be implemented in other types of farm (e.g., small/family-run breeding farms, big breeding farms working independently) to better collect data on general and specific critical points of biosecurity and to implement critical self-reflection and forms of adjustment for practical purposes. Lastly, another development of this project could be the promotion of participatory processes with public and private veterinarians to foster negotiation, sharing, and co-definition of sustainable, effective, long-term objectives and solutions to address future epidemic emergencies.

Ethics in publishing

The article reports data on a social research study based on the perception and opinion of breeeders, no experimentation was carried out on humans or animals. To comply with the General Data Protection Regulation (EU) 2016/679, a privacy agreement was requested to all participants involved in the research activities and a paper document was signed by breeders before the interview began.

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CRediT authorship contribution statement

Conceptualization, (Stefania Crovato), (Anna Rosa Favretto), (Giandomenico Pozza), (Giulia Mascarello), (Francesca Zaltron); Data curation, (Stefania Crovato), (Giulia Mascarello), (Anna Rosa Favretto), (Francesca Zaltron); Investigation, (Francesca Zaltron), (Anna Rosa Favretto); Methodology, (Stefania Crovato), (Anna Rosa Favretto) and (Francesca Zaltron); Project administration, (Giandomenico Pozza), (Stefania Crovato) and (Giulia Mascarello); Supervision, (Paolo Mulatti), (Tiziano Dorotea); Writing—original draft, (Stefania Crovato), (Alessio Menini); Writing—review & editing, (Stefania Crovato), (Alessio Menini), (Giulia Mascarello), (Paolo Mulatti), (Tiziano Dorotea). All authors have approved the final article. All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

none.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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